Prerequisites

Before shippers can start booking dock appointments, they need to take some settings in other apps, such as the **Manage Locations** or **General Settings** app.

You must create a location master data in the **Manage Locations** app. For more information on creation of a location, see Creating a Location

Settings to Enable Appointment Booking

To book appointments with the **Manage Dock Appointments** app and to enable the carriers to self-book dock appointments, you must enable certain settings. The carrier can create a dock appointment before or after confirming a freight order based on your settings.

To enable you and the carrier to book dock appointments for a confirmed freight order, you must perform the following:

1. In the **Manage Locations** app, you must create a location with the following settings:
    - Specify **Object Type** as **Business Partner**.
    - Specify the source system ID as it appears in your SAP Transportation Management (SAP TM) system.

      Note that the format must be specified exactly as it appears in the SAP TM system. For example, QM7_910.
    - In the **External ID** field, enter the business partner ID of the business partner associated with the purchasing organization as it appears in the SAP Transportation Management system.

      Note that in the **General Settings** app, the system represents the ordering party ID in the format <source system ID>:<external ID>. For more information, see Creating a Location.
2. In the **Docking Location Settings** app, you must maintain this location as a loading or unloading point and as an active location. For more information, see Manage Entities.

You can also enable a carrier or a warehouse operator to create an appointment for a freight order that has not yet been confirmed. To do so, along with the above-mentioned settings, in the **General Settings** app, you must perform the following:

1. In the **Define Appointment Settings** tab, add the ordering party by selecting the entry with the system ID and external ID that you created.
2. For this ordering party, under the **Maintain Appointment** setting, enable the option **On Freight Order Creation** button.

Enabling Role-Based Access

Role-based access configuration for the Manage Dock Appointments app involves creating a role from a role template and setting specific location attributes.

To enable role-based access of **Manage Dock Appointments** app and **Manage Gate Operations** app, proceed as follows:

1. Create a role using the AttributedLocationTemplate role template.

   You can find this role template under the lbn-live!t1590 application identifier.
2. For the newly-created role, set the location_id attribute in AttributedLocationTemplate to <Source System>:<External ID> of the docking location that you want the shipper to access, for example *QM7_910: LBN_WDF*.

   You can find the Source System and External ID of a location in the **Manage Locations** app.
3. Add this role to the role collection that has been assigned to the shipper.

Result

In the **Manage Dock Appointments** app, shippers who have the AttributedLocationTemplate role template assigned can only access the dock appointments in the docking location that matches the location_id attribute set in AttributedLocationTemplate.

In the **Manage Gate Operations** app, shippers who have the AttributedLocationTemplate role template assigned can only access gate logbooks associated to the yard location that matches the location_id attribute set in AttributedLocationTemplate.

ℹ Note

- In the **Manage Dock Appointments** app, shippers who do not have the AttributedLocationTemplate role template assigned can access all docking locations.
- In the **Manage Gate Operations** app, shippers who do **not** have the AttributedLocationTemplate role template assigned can access all gate logbooks.

Integration of Dock Appointment Scheduling with Freight Orders

Integration of dock appointment scheduling with freight orders in the transportation management (TM) system streamlines logistics by automatically creating and updating appointments based on changes in freight order details.

The dock appointment scheduling functionality is integrated with the freight orders from your external systems such as your transportation system. The system uses an inbound message from your transportation system to create a freight order for the carrier in your SAP Business Network for Logistics system. If you have maintained the relevant settings for the carrier in the **General Settings** app, the system created appointment requests for this freight order. For more information on the settings, see General Settings.

Confirming a Freight Order

When a carrier confirms a freight order using the **Freight Orders for Confirmation** app in SAP Business Network for Logistics, the system creates appointment requests for this freight order.

When you make any changes to a confirmed freight order in your transportation system or in SAP Business Network for Logistics, the system updates the appointments accordingly.

The following table describes what happens when you update a freight order:

Freight Order Update Type	Appointment Status	Notification
Change of arrival or departure dates	System updates the open and provisional appointments System cancels the confirmed appointments **Exception**: If the changed dates match the start and end date of the scheduled appointment, the appointment will not be canceled.	System triggers notification on cancellation
Change of docking location or loading point	System updates the open and provisional appointments System cancels the confirmed appointments	System triggers notification on cancellation
Change of document references	No change in status. System updates the document references.	

Freight Order Update Type	Appointment Status	Notification
Other changes	No change in status. System updates the changes.	

Canceling or Rejecting a Freight Order

When you cancel a freight order from your transportation system or when a carrier rejects a freight order from the **Freight Orders for Confirmation** app, the system cancels the relevant appointment requests and the booked appointments. The appointments will not be visible on the **Self-Book Dock Appointments** app for the carrier and the **Manage Dock Appointments** app.

In addition, an appointment is automatically canceled if one of the following is changed:

- Arrival dates
- Departure dates
- Loading point
- Docking location

ℹ Note

If the changed dates match the start and end date of the scheduled appointment, the appointment will not be canceled.

If the start and end date of the scheduled appointment is within the range of the new acceptable start and end date, the appointment will also not be canceled.

Reconfirming a Freight Order

When you reconfirm a canceled freight order from your transportation system or when a carrier reconfirms a rejected freight order from the **Freight Orders for Confirmation** app, the system reopens the relevant appointment requests for booking.

Preferred Slots / Slots Preferred by Shipper

If the shipper provides the relevant acceptable and requested date and time information, it will be displayed when you maintain an appointment.

- The requested dates correspond to the **Loading of Cargo Start Date/Time** and **Loading of Cargo End Date/Time** fields of the respective stop in the transportation management (TM) system.

  The UI fields are related to the `TransportationOrderRequest` message type in TM where start date time and end date time are inside the `RequestedLoadingDateTimePeriod` and `RequestedUnloadingDateTimePeriod` tags.

- The acceptable dates correspond to the **Start Date/Time Confirmed by Carrier** and **End Date/Time Confirmed by Carrier** fields of the respective stop in the transportation management (TM) system.

  The UI fields are related to the `TransportationOrderRequest` message type in TM where start date time and end date time are inside the `AcceptableLoadingDateTimePeriod` and `AcceptableUnloadingDateTimePeriod` tags.

Additional Address to Fix Loading Point

The subcontracting process passes the loading point ID and unloading point ID from the transportation management system (TM) to SAP Business Network for Logistics. This is done through the **Location** field under **Additional Address at Source Location** or **Additional Address at Destination Location** in TM, which is sent through the `InternalID` fields under

`DetailedLoadingLocation` or `DetailedUnloadingLocation` in the *Provider API to Create Freight Order for Subcontracting* (`TransportationOrderRequest_Out`).

If the internal ID matches an existing loading point ID, the system preselects and fixes this loading point. It then cannot be changed by the shipper or carrier in SAP Business Network for Logistics.

Manage Entities

Manage Entities app allows users to define and manage attributes for loading points, docking locations, gates, and yards.

You can use this app to define and view the following settings:

- **Loading Point**

  You can define the attributes of a loading point and manage the business hours and block time.

- **Docking Location**

  The loading points situated in the same geographical location are grouped into one docking location. You can define the attributes of a docking location and manage the corresponding loading points.

- **Gate**

  You can define the attributes of a gate and manage the business hours.

- **Yard**

  You can define the attributes of a yard and maintain the corresponding gates and docking locations.

ℹ Note

To use this app, you must add one or several roles to your role collection. The required role templates and application identifiers are described in the Available Roles for Shippers section of the Administration Guide for Shippers.

Defining Loading Point Settings

Loading point settings define operational parameters such as business hours, capacity, and time slot booking for efficient management of loading and unloading activities.

To define the settings for a loading point, proceed as follows:

1. Choose the **Loading Point** tab on the left pane.
2. Choose the **Add Loading Point** button.

   The create loading point pane appears on the right.
3. Define settings for the following sections:
   - **Details**
   - **Business Hours**
   - **Block Time**
4. Choose **Save** to save the loading point as a draft. Choose **Save and Activate** to activate the loading point settings.

ℹ Note

The following fields cannot be changed any longer after the loading point was activated:

Under **Location Details**:

- Loading Point ID
- Location ID
- Description
- Address
- Capacity
- Activity Type

Under **Time Details**:

- Time Zone
- Time Slot Booking
- Time Slot Duration

Details

Enter the following information in the **Details** section:

- **Loading Point ID**

  Enter the loading point ID.

  The loading point ID is the identification of a single loading point for differentiation. You must always book a dock appointment for a specific loading point. That's why you need to create a loading point. A loading point can have several areas (for example, doors), which are represented by the capacity. You must always assign the loading point to a docking location in order to select a loading point for appointment booking.

- **Location ID**

  Select a location ID (optional).

  You maintain location master data in the **Manage Locations** app. The location ID is composed of the source system and the external ID as follows: *app. The location ID is composed of the source system and the external ID as follows:<Source System>:<External ID>*. It is retrieved from the location master data one time at the beginning. After activation it cannot be changed anymore even if the values in the **Manage Locations** app change.

  > **Note**
  >
  > The location ID of the loading point is not used to determine the loading point of the appointment. Instead, the location ID of the docking location is used to determine the docking location for appointment booking.

- **Capacity**

  Specify the number of loading/unloading activities that can be performed concurrently at the loading point.

  In a logistics setting, a capacity value of 10 typically indicates that there are 10 doors available in the loading area, allowing 10 trucks to be loaded or unloaded simultaneously. However, the exact meaning of capacity depends on how your operational entities – such as loading points and docks – are defined and modeled within your system. For instance, if each door is represented as a loading point in your configuration, then the capacity assigned to each loading point is 1.

- **Address**

- **Activity Type**

  Specify whether you can perform loading, unloading, or both activities in this loading point.

- **Time Zone**

- **Appointment Date Change**

  If you enable this indicator, the carriers can change the date of a dock appointment. You can specify a range for the loading point so that carriers can change the date within the specified range. Note that a date change is only possible before the cut-off time.

  - **Advance Up To**

    Specify the number of days up to which the carrier can advance the dock appointment date.

  - **Postpone Up To**

    Specify the number of days up to which the carrier can postpone the dock appointment date.

  Note

  This setting that you maintain for a loading point, takes precedence over the settings you specify for the **Enable Appointment Date Change** in the docking location.

- **Time Slot Booking**

  Time slot booking is disabled by default so that you cannot specify a time slot duration. While booking an appointment, you do not have to select a time slot and the appointment is valid at any time of the day of the appointment date. The time slot duration is set to *0*.

  If you enable time slot booking for this loading point, you can define the time slot duration for which this loading point can be booked.

  - **Time Slot Duration**

    You can select a time slot duration for which the loading point can be booked. You can enter a duration between 5 minutes and 23:59 hours.

  - **Restrict Maximum Duration of Consecutive Time Slots**

    You can define whether the maximum duration of all consecutive time slots is restricted. This setting ensures that the sum of time slots to be booked by your carrier cannot exceed the maximum consecutive duration of all adjacent time slots. This restriction only affects the booking for the carrier when they select more than one time slot. They can always book at least one time slot. As the shipper, you can always exceed the maximum duration.

    For more information, see the following blog: SAP Business Network for Logistics - Dock Appointment Scheduling: Max. Consecutive Timeslot Duration.

  - **Maximum Consecutive Duration**

    You can define the maximum duration of all adjacent time slots.

  - **Use Derived Appointment Duration**

    If you enable the **Use Derived Appointment Duration** toggle, the system forces carrier users to book appointment durations based on input from an external system.

    1. This can, for example, come from the `TransportationOrderRequest` message type in SAP Transportation Management or transportation management (TM) in SAP S/4HANA, using the `RequestedLoadingDateTimePeriod` and/or `RequestedUnloadingDateTimePeriod` fields.

    2. Additionally or instead, you can add a custom implementation and provide the `LoadingDuration` and `UnloadingDuration` fields within the `TransporationStage`. As a prerequisite, you must implement the

following SAP Notes:

- 3680384 - Scheduling Settings Simulation Interface
- 3662698 - Loading and Unloading Duration in TransportationOrderRequest_Out
- 2474555 - Additional SOA Documentation for Enterprise Services "Transportation Order"

If both options are implemented, the `LoadingDuration` and `UnloadingDuration` fields take precedence.

3. Another option is through the *Provider API for Dock Appointment Scheduling* (`GenericAppointment_In`), using the `RequestedStartDateTime` and `RequestedEndDateTime` fields.

For more information about the setup and the underlying processes, see the following blog: SAP Business Network for Logistics - Dock Appointment Scheduling: Use Derived Appointment Duration.

- **Overbooking**

  You can enable or disable overbooking for this loading point. When you enable overbooking, the system allows the user to book appointments even after the available slots are booked. If you enable overbooking, the system displays the following:

  - **Overbooking Limit**

    If you enable the overbooking setting, you must specify the overbooking limit. Overbooking limit refers to the number of appointments that can be booked once the available slots are booked. Note that as of now you can set the overbooking limit up to 50.

  - **Overbooking for Carriers**

    If you enable the overbooking setting, you must either enable or disable the overbooking option for the carriers. When you enable the overbooking option for carriers, the carriers can book appointments even after the available slots are booked. Note that the overbooking limit that you set here applies for the carriers as well when you enable this setting

- **Capacity and Time Slot At**

  You can specify the level at which you want to enable the capacity and time slot settings. You can select one of the following options:

  - **Loading Point Level**
  - **Validity Period Level**
  - **Working Hours Level**

  Note

  After loading point activation, this field can only be changed to a lower level (**Loading Point** > **Validity Period** > **Working Hours**), and the settings defined for the newly selected level will be used. It can't be changed back to a higher level.

  Note

  The Microsoft Excel upload feature only works if you have selected the **Loading Point Level** option.

  Based on the setting you specify here, you can maintain the capacity and time slot at the respective levels.

- **Assignment Details**

  - **Assignment with Docking Locations**

    If you enable this setting, the loading point can be associated with multiple docking locations in the same time zone. If you disable this setting, the loading point can be associated with only one docking location in the same time zone.

  - **Assigned Docking Locations**

    You can select the docking locations in the same time zone with which you want to share the loading point.

After selecting the relevant docking locations, you can assign the loading point to these additional docking locations. To do so, you need to edit the docking location, choose the + icon, and select the loading point from the list under **Add Loading Point**.

- **Remarks**

  You can enter the additional information about the loading point in this section.

Validity Period

Choose the add icon in this section to add an entry of validity period.

> **Note**
> You can add multiple entries of validity period per working day for a loading point. However, the combination of period and working days specified in an entry of validity period must not overlap with the consecutive entries.

Enter the following information:

- **Period**

  Specify the duration for which the entry of validity period is valid.

- **Working Days**

  Specify the working days of the loading point.

- **Working Hours**

  Specify the working hours for the specified working days.

  To add another entry of validity period, choose the add icon. You can create up to 10 entries per validity period.

  It is recommended that you specify the working hours in accordance with the time slot that you have maintained for this loading point.

Deleting Validity Periods

To delete an entry of validity period, choose the delete icon that appears next to the entry. You can also choose **Delete** from the details pane that appears on the right when you select the entry of validity period.

Note that you cannot delete an entry of validity period if the entry contains appointments in the future.

You can only delete an entry that either does not contain any future appointments or contains appointments in the past.

Modifying Validity Periods

You can select an entry of validity period and modify the details on the right pane.

Modifying Periods

You can increase or reduce the period.

> **Note**
> You can only reduce a period that does not contain any future appointments or contains appointments that are in the past.
>
> You cannot reduce the period, if the period that you are reducing contains appointments that are in the future.

If you have enabled **Capacity and Time Slot At** as **Validity Period Level**, you can view and modify the **Capacity** and **Time Slot Duration** in this pane.

Modifying Working Days

You can add a working day for the selected entry of validity period.

Note that the system allows you to delete a working day for a time period only when the deleted working day does not contain any future appointments.

Modifying Working Hours

You can increase or reduce the working hours.

> **Note**
>
> When you modify the working hours, there might be inconsistencies in the start and end time of the time slots. Hence, it is recommended that you modify the working hours in multiples of the time slot duration.
>
> You cannot modify the working hours, if the selected entry of validity period contains appointments that are in the future.
>
> If you have enabled **Capacity and Time Slot At** as **Working Hours Level**, you can view and modify the **Capacity** and **Time Slot Duration** in this pane.

Block Time

You can block time for special occasions, such as public holidays, maintenance, or cleaning.

Choose the add icon in this section to add a time block. The system displays the block time pane on the right.

- **Description**

  Specify a description for the time block.

- **Disable Appointment Booking**

  Specify whether you want to disable appointment booking only for the carriers or for all.

  If you select **For All**, both you and the carriers cannot book appointments during the blocked time.

  If you select **For Carriers**, only the carriers cannot book an appointment during the blocked time. The blocked time will be available for you to book an appointment.

- **Max. Capacity**

  When you enable the **Max. Capacity** toggle, the maximum capacity is blocked for this loading point, and the capacity in the dropdown above is disabled because it is no longer relevant.

  For more information, see the following blog: SAP Business Network for Logistics - Dock Appointment Scheduling: Maximum Capacity Blocker.

- **Including Overbooking**

  When you enable the **Include Overbooking** toggle, the capacity maintained under **Overbooking Limit** is blocked for this loading point. You can only enable this toggle when the **Max. Capacity** toggle is enabled.

- **Capacity**

  In the dropdown menu, you can view the capacity that you have specified for this loading point. You can select a capacity that you want to block from the dropdown menu.

- **Date and Time**

  You can specify the start and end date and time, and you can choose between single blocks and recurring blocks.

- **Single Block**: Creates a single occurrence of the time block.

- **Recurring Block**: Creates multiple occurrences based on the specified **recurrence interval** and the **recurrence end date**.

  You can specify a daily, weekly or monthly recurrence interval, such as every third week on a specific day of the week.

  For monthly recurrence intervals, you can choose between the following options:

  - **Day**: A specific day of the week, such as every first Monday of the month.
  - **Date**: A calendar date, such as **1** for the first day of the month.

  For more information, see the following blog: SAP Business Network for Logistics - Dock Appointment Scheduling: Announcing Recurring Block Times.

- **All Day Event**

  When you enable the **All Day Event** toggle, you can create a block for an entire day or for several days in a row.

  Note: When creating an **All Day Event**, ensure that the duration of the appointment is less than the chosen recurrence interval. For instance, if your appointment lasts two days, the earliest possible recurrence is every three days or more. This means the recurrence interval must always be longer than the appointment duration.

- **Start Date and Time**

  Specify the start date and time for the time block.

- **End Date and Time**

  Specify the end date and time for the time block.

- **View History**

  When you enable the **View History** toggle, the system also displays the appointments that are in the past. By default, time blocks in the past are hidden.

You can block the time for a loading point that has been shared with multiple docking locations. When you do so, the system blocks the entire capacity of the loading point for the blocked period across all the docking locations with which the loading point has been shared.

ℹ Note

Duration

The maximum duration of a single block is 31 days.

If you want to block your loading point for more than 31 days, you can use the **Deactivate** option of the loading point or use a recurring block. For more information, see the status **Inactive** under **Header** on this page.

The maximum duration of a recurring block depends on the recurrence pattern:

- **Daily**: The maximum duration is 31 days, and it can occur up to every 31 days. The duration must be shorter than the recurrence frequency. The recurrence end date can be set up to one year from the start date.
- **Weekly**: The maximum duration is 24 hours, and it can occur up to every 6 weeks. The recurrence end date can be set up to one year from the start date.
- **Monthly**: The maximum duration is 24 hours, and it can occur up to every 12 months. It must be shorter than the recurrence frequency. The recurrence end date can be set up to three years from the start date.